

Per-Operator Attribution of the Fake-Quantization-to-Integer Gap in INT8 Inference

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Abstract—INT8 deployment decisions are made on a simulator. The network runs in floating point with quantize–dequantize (“fake-quantization”) nodes inserted, and its accuracy stands in for the accuracy of the integer target. We measure how far that stand-in holds — for accuracy and for tensors separately — against an integer engine that is bit-exact with the TensorFlow Lite reference kernels. On six networks, two quantization schemes and complete test splits, it holds for accuracy: every fake-versus-integer difference is within 0.20 pp and all twelve 95% confidence intervals contain zero. It fails for tensors: 28.5–72.7% of the output codes differ, on 95–100% of the images. Teacher forcing attributes the divergence per operator. Across these networks convolution, linear and pooling each disagree on under 1% of their own codes, while integer LayerNorm disagrees on 24.8%. Making the simulator integer-faithful at LayerNorm removes that source entirely (24.77% $\rightarrow$ 0.00%) but moves the propagated output mismatch only from 72.7% to 65.7%: a simulator cannot be repaired into a source of bit-accurate vectors one operator at a time. We also price a requantization choice the accuracy proxy cannot see — replacing rounding with truncation costs 0.90–3.37 pp in all 18 (model, seed, mode) combinations, and the sign of that cost transfers across seeds while its magnitude does not. The decoupling survives the removal of the argmax: on a dense-output network the quality metric is faithful to 6×10^{-4} dB while 6.3% of the delivered pixel codes differ, and tripling that network’s operator count at a fixed width raises the differing fraction to 69.2% with the metric still faithful.

Index Terms—Quantization, integer arithmetic, embedded inference, neural processing units, reproducibility.

I. INTRODUCTION

TWO programs decide an INT8 deployment. The one that answers questions is a floating-point graph with quantize–dequantize (“fake-quantization”) nodes, whose accuracy is reported as the quantized accuracy [1], [2]. The one that ships is an integer program: int8 weights, an int32 accumulator, and a fixed-point multiply-and-shift requantization [3]. Practice treats the two as interchangeable, and for accuracy they largely are [4].

Accuracy is not all that is asked of the simulator. Silicon bring-up and compiler validation need *golden vectors*: the exact integer tensors the device must produce. Triaging a post-training quantization regression needs to know *which* operator started the divergence. Both are tensor questions, and to our knowledge neither has been measured. We ask three. **RQ1**: does agreement on accuracy imply agreement on tensors? **RQ2**: which operators create the divergence, and does removing the largest one close it? **RQ3**: what does the accuracy proxy fail to price?

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Code, checkpoints and every result file needed to reproduce the tables and figures: <https://github.com/sokldjs554/npuloop>.

Answering them needs a decomposition that current layer-wise error analyses do not provide: separating what an operator’s own integer arithmetic contributes (*local* mismatch, isolated by teacher forcing) from what survives end-to-end execution (*propagated* mismatch). That decomposition, reported with paired standard errors on full test splits, is this letter’s methodological contribution; Findings 1–6 are its empirical ones.

II. BACKGROUND AND RELATED WORK

A fake-quantization graph rounds and clips a tensor to the quantization grid and then returns to float, so every operator still runs in float32. The integer program instead keeps int8 codes end to end and rescales with a Q31 multiplier and a shift [1], [3]. The two therefore agree on *intent* and are free to disagree on every least-significant bit.

MQBench [5] measures the gap between academic quantization and deployable backends and reports it as accuracy; the quantization white papers [4], [2] treat fake quantization as the simulation of the integer pipeline; layer-wise error analyses in deployment toolchains quantify how far a quantized layer is from its *float* counterpart. Ours is a different measurement: how far the *simulator* is from the *integer execution*, in codes rather than in signal-to-noise ratio. Integer-only transformer inference has established integer LayerNorm and softmax kernels [6], [7]; our contribution there is not the kernel but the size of its disagreement with the simulator. Accumulator width has been studied as a training-time constraint [9]; the axis here is instead the rounding of the requantization step, a pure implementation choice.

III. MEASUREMENT SETUP

A. Integer engine and external validation

A calibrated fake-quantization graph is exported to an integer graph: int8 weights, int32 biases, and a per-output-channel Q31 multiplier and shift. Execution follows the `gemmlowp/TensorFlow Lite` semantics of `MultiplyByQuantizedMultiplier` [3]. Two independent implementations (NumPy and C++) must agree on *every* intermediate tensor, and a third, Python-free C++ executor reproduces the same codes from a serialized file.

We validate the engine against an external oracle, not against itself. Building the integer graph from a TensorFlow Lite full-integer model’s own scales, zero-points, weights and biases, the two agree on all tensors for 1000/1000 images when convolution and pooling use the legacy double rounding and the fully-connected layer uses single rounding. For calibration

TABLE I

FAKE QUANTIZATION VERSUS BIT-EXACT INTEGER EXECUTION, COMPLETE TEST SPLITS. PC: PER-CHANNEL WEIGHTS, PT: PER-TENSOR. Δ IS (INTEGER – FAKE) TOP-1 ON THE SAME IMAGES, WITH ITS PAIRED STANDARD ERROR. “CODES” IS THE FRACTION OF OUTPUT LOGIT CODES THAT DIFFER; “IMGS” THE FRACTION OF IMAGES WITH AT LEAST ONE DIFFERING CODE.

Network	Sch.	float32	fake	integer	Δ (pp)	95% CI	agree (%)	codes (%)	imgs (%)
ResNet-20 ReLU	PC	89.59	89.54	89.55	$+0.01 \pm 0.08$	$[-0.14, +0.16]$	99.37	50.1	99
ResNet-20 ReLU	PT	89.59	89.61	89.67	$+0.06 \pm 0.08$	$[-0.11, +0.23]$	99.18	51.7	99
ResNet-20 SiLU	PC	90.34	90.35	90.26	-0.09 ± 0.09	$[-0.26, +0.08]$	99.06	61.6	100
ResNet-20 SiLU	PT	90.34	90.17	90.26	$+0.09 \pm 0.09$	$[-0.08, +0.26]$	99.07	62.9	100
MobileNetV2-0.5	PC	90.23	90.30	90.29	-0.01 ± 0.08	$[-0.16, +0.14]$	99.28	50.4	100
MobileNetV2-0.5	PT	90.23	90.23	90.27	$+0.04 \pm 0.07$	$[-0.10, +0.18]$	99.37	51.4	100
ViT-128/6	PC	80.98	80.87	80.97	$+0.10 \pm 0.13$	$[-0.15, +0.35]$	97.93	72.7	100
ViT-128/6	PT	80.98	80.77	80.97	$+0.20 \pm 0.12$	$[-0.04, +0.44]$	98.07	72.6	100
Inception-32	PC	89.59	89.55	89.56	$+0.01 \pm 0.05$	$[-0.09, +0.11]$	99.69	28.5	95
Inception-32	PT	89.59	89.52	89.47	-0.05 ± 0.06	$[-0.17, +0.07]$	99.52	31.1	96
ResNet-20 (Imagenette)	PC	83.97	84.05	83.97	-0.08 ± 0.10	$[-0.27, +0.12]$	99.52	35.7	95
ResNet-20 (Imagenette)	PT	83.97	84.00	84.05	$+0.05 \pm 0.11$	$[-0.17, +0.27]$	99.31	36.2	96

of how tight that criterion is: TensorFlow Lite’s own optimized delegate and its reference kernels differ on 155/1000 images of the same model.

B. Local and propagated mismatch

For one image, let f_n be the codes the fake-quantization graph produces at node n (its quantizer output, converted to integer codes) and i_n the codes the integer engine produces when run end-to-end. The *propagated* mismatch at n is the fraction of elements with $f_n \neq i_n$. The *local* mismatch feeds node n ’s integer kernel the fake-quantization codes of its inputs (teacher forcing) and compares only its own output. Local mismatch isolates what an operator’s integer arithmetic contributes; propagated mismatch shows how far that contribution travels.

C. Statistics, models and schemes

Accuracy differences are paired: the same images are evaluated three times (float, fake-quantization, integer). We report the mean per-image difference in correctness with its standard error $s/\sqrt{n}$ and a 95% Wald interval. Accuracy columns use the complete test split (10,000 images for CIFAR-10, 3,925 for Imagenette); the per-node decomposition uses a fixed batch of 250 images (125 for Imagenette).

Five CIFAR-10 networks — ResNet-20 with ReLU and with SiLU, MobileNetV2-0.5 (depthwise), a concatenation-branch Inception-style CNN, and a ViT-128/6 (12 attention blocks, 13 LayerNorms) — plus two Imagenette networks at 128×128 : a ResNet-20 and the super-resolution network of Section IV-B. Two post-training schemes, per-channel weights with asymmetric activations (PC) and per-tensor weights (PT), both calibrated on 512 training images. Everything runs on CPU; every number below is emitted by a script from a committed results file.

IV. RESULTS

A. RQ1: accuracy agrees, tensors do not

Table I is the main measurement. The largest fake-versus-integer accuracy difference is 0.20 pp, the largest $|\Delta|/\text{SE}$ is 1.6, and all twelve confidence intervals contain zero. At these

TABLE II

DENSE OUTPUT: ESPCN $\times 2$ ON IMAGENETTE, 3,925 IMAGES OF 49,152 OUTPUT VALUES EACH, AT TWO DEPTHS OF ONE WIDTH PROFILE. PSNR IN dB ON $[0, 1]$. QUANTIZATION ITSELF COSTS 0.28–0.67 dB AT 3 CONVOLUTIONS AND 0.57–1.15 dB AT 9, THREE ORDERS OF MAGNITUDE MORE THAN THE SIMULATOR-TO-INTEGER DIFFERENCE BESIDE IT, WHILE THE FRACTION OF DIFFERING CODES MOVES WITH DEPTH BY AN ORDER OF MAGNITUDE.

Convs	Sch.	float32	fake	integer	Δ (10^{-3} dB)	codes (%)	max $ \Delta c $
3	PC	30.234	29.956	29.955	-0.64 ± 0.04	6.3	3
	PT	30.234	29.567	29.568	$+0.27 \pm 0.06$	18.2	3
9	PC	30.113	29.544	29.543	-0.51 ± 0.18	69.2	10
	PT	30.113	28.966	28.968	$+2.88 \pm 0.19$	73.0	10

sample sizes the accuracy proxy is not distinguishable from exact, and image-level top-1 agreement is 97.9–99.7%. The same rows disagree on tensors: between 28.5% (Inception-32) and 72.7% (ViT-128/6) of the output logit codes differ, on 95–100% of the images. The transformer is the extreme case but not a special case — the CNNs sit at 50–63% and the larger-input Imagenette model at 36%.

Finding 1. Accuracy agreement and tensor agreement are decoupled by more than two orders of magnitude, on every network and both schemes.

Implication. A reported 99% accuracy agreement is no evidence that the simulator can supply a verification vector.

B. RQ1 where the tensor is the deliverable

Every network above is a classifier, so an argmax stands between the output tensor and the answer. We remove it with a sub-pixel super-resolution network [8] — $64 \rightarrow 128$ on Imagenette, 30.23 dB against 27.26 dB for a nearest-neighbour upsampler — whose output is the picture. The pixel shuffle needed no new operator: it is a reshape and a transpose. To separate depth from task we measure two depths at one width profile, three convolutions and nine; the deeper one is not the better one (30.11 dB), only the same task at three times the operator count.

Table II repeats the pattern and puts a number on depth. At three convolutions quantization costs 0.279 and 0.667 dB and the simulator predicts that cost to within 6×10^{-4} dB —

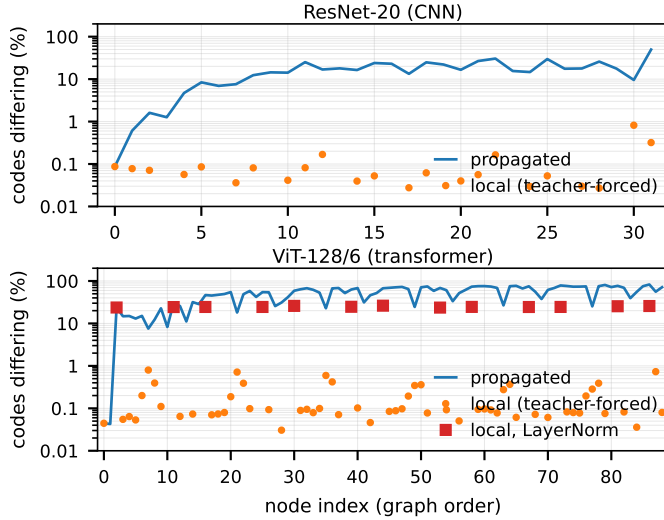


Fig. 1. Per-node code mismatch, PC scheme. Local (teacher-forced) mismatch is below 1% for every operator except LayerNorm; propagated mismatch rises from the first convolution to 50–73% at the output. Log scale; nodes with zero local mismatch are not drawn — every lookup-table activation, and all but five of the 100 residual additions.

resolvable here ($|\Delta|/\text{SE} = 17.6$), three orders of magnitude below the cost predicted — yet 6.3% and 18.2% of the output pixel codes differ, on all 3,925 images, by up to three levels. At nine the metric stays faithful to 3×10^{-3} dB while the differing fraction rises to 69.2% and 73.0% and the largest deviation to ten levels. The per-operator structure is the CNNs’ at both depths: convolution disagrees locally on 0.06–0.40% of its codes, always by ± 1 LSB, first at the first convolution.

Finding 2. The decoupling is not an artifact of classification, and depth rather than task sets the size of the propagated fraction: nine convolutions on the same images move it from 6.3% to 69.2%, into the 28–73% band the classifiers occupy, the quality metric faithful throughout.

Implication. A simulator can no more supply a reference image than a reference logit vector, and the deeper the graph the less of the delivered tensor it reproduces.

C. RQ2: where the divergence is born

Fig. 1 separates the two components. Local mismatch is small and flat: convolution disagrees on 0.02–0.93% of its codes, linear on 0.03–0.96%, average pooling on 0.1–2.7% (rounding ties), and the int8 lookup-table activations on exactly 0%. Every one of these is a ± 1 least-significant-bit disagreement — as is every local disagreement in the study, LayerNorm’s included. Propagated mismatch instead rises monotonically from the first convolution — the first divergence is that node in all twelve rows — and reaches 28–73% at the output.

One operator is not small. Integer LayerNorm disagrees with its floating-point counterpart on 24.8% of its codes, with a maximum of 26.2% over the 13 instances. The integer implementation normalizes with an int64 sum and an exact integer square root, while the simulator computes in float32 and then snaps to the grid, so roughly one element in four lands on the other side of a rounding boundary.

TABLE III

MAKING THE SIMULATOR INTEGER-FAITHFUL AT LAYER NORM (ViT-128/6, 10,000 IMAGES). THE INTEGER PROGRAM IS UNCHANGED BETWEEN THE TWO ROWS OF EACH SCHEME; ONLY THE SIMULATOR’S LAYER NORM DIFFERS. THE MEAN ABSOLUTE LOGIT DIFFERENCE FALLS FROM 0.119 TO 0.091 (PC) AND FROM 0.118 TO 0.093 (PT).

Sch.	LayerNorm	local (%)	codes (%)	agree (%)	Δ (pp)
PC	float32	24.77	72.7	97.93	$+0.10 \pm 0.13$
PC	integer (13)	0.00	65.7	98.29	$+0.25 \pm 0.11$
PT	float32	24.76	72.6	98.07	$+0.20 \pm 0.12$
PT	integer (13)	0.00	66.0	98.60	$+0.13 \pm 0.10$

TABLE IV

ROUNDING-MODE COST AGAINST THE REFERENCE IMPLEMENTATION, MEAN $\pm$ STANDARD DEVIATION OVER THREE SEEDS (PP).

Mode	ResNet-20 ReLU	ResNet-20 SiLU	MobileNetV2-0.5
single	-0.03 ± 0.09	$+0.01 \pm 0.02$	-0.00 ± 0.13
half_even	-0.07 ± 0.09	$+0.02 \pm 0.09$	$+0.00 \pm 0.09$
truncate	-0.98 ± 0.12	-2.31 ± 0.85	-2.12 ± 0.21
floor	-1.42 ± 0.06	-2.42 ± 1.04	-1.43 ± 0.40

Finding 3. Outside LayerNorm, every operator’s own integer arithmetic disagrees on under 1% of its codes, and every such disagreement is ± 1 LSB. The end-to-end gap is therefore grown by propagation, not contributed by any one kernel.

Finding 4. Integer LayerNorm is the single large local source, two orders of magnitude above every other operator.

Implication. When porting a transformer to INT8, suspect LayerNorm first and expect ± 1 LSB everywhere else.

D. RQ2: removing the largest local source

If the divergence were a sum of per-operator errors, removing the largest would help proportionally. We test this: the simulator’s LayerNorm is replaced by a module that computes the engine’s arithmetic on the input codes, bit-exact by construction, while the exported integer program is unchanged (the experiment asserts so).

Table III shows the outcome. Local LayerNorm mismatch goes from 24.77% to exactly 0.00%, and the mean absolute logit difference improves by about 23%. The propagated output-code mismatch moves only from 72.7% to 65.7% (per-tensor: 72.6% $\rightarrow$ 66.0%). Accuracy moves within about two paired standard errors in either direction and cannot be read as an improvement.

Finding 5. Eliminating the single largest local source recovers about a tenth of the propagated gap, because what remains is the amplification of ± 1 LSB events from operators that individually disagree on under 1% of their codes.

Implication. Golden vectors have to come from an integer implementation validated against an external reference. Repairing a floating-point simulator operator by operator converges too slowly to substitute for one.

E. RQ3: a cost the accuracy proxy hides

The requantization multiply-and-shift ends in a rounding step, which an implementation may be tempted to replace with

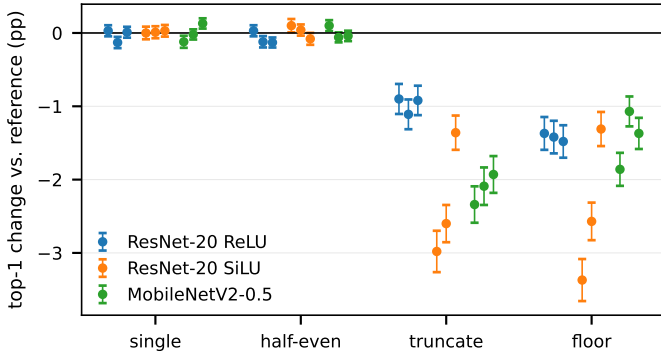


Fig. 2. Top-1 change against the reference requantization rounding, complete test split, three independently trained checkpoints per network (error bars: paired standard error). The biased modes lose in every combination; how much they lose varies by seed.

a plain arithmetic shift. Fig. 2 and Table IV price that choice on full test splits, with three independently trained checkpoints per network.

`truncate` and `floor` are worse than the reference in all 18 (model, seed, mode) combinations, by 0.90–3.37 pp, with $|\Delta|/SE$ between 4.4 and 11.8; the alternative single-rounding and round-half-to-even paths stay within noise ($|\Delta| \leq 0.13$ pp). The magnitude does not transfer. For ResNet-20 SiLU the cost of `truncate` ranges from 1.36 to 2.98 pp across three seeds, a seed standard deviation equal to 37% of the effect, while ResNet-20 ReLU stays within a tenth of its effect. Which biased mode is worse is a property of the network, not of the arithmetic: `floor` loses more on all three ResNet-20 ReLU seeds, `truncate` on all three MobileNetV2 seeds.

Finding 6. The sign and significance of a rounding-mode cost are seed-robust; its magnitude is not, varying by up to 37% of the effect across three seeds of the same architecture.

Implication. Do not quote a single-seed number as the cost of a rounding mode.

V. THREATS TO VALIDITY

External. Beyond the CIFAR-10-scale and Imagenette networks we trained, the split holds on a ResNet-50 carrying Keras’s ImageNet weights (53 convolutions, $K = 4608$): accuracy within $|\Delta|/SE \leq 0.2$ while 57–58% of output codes differ on all 3925 images — an ImageNet-scale graph and ImageNet weights, not the ImageNet-1k validation set. Depth is the one scale axis varied directly (Section IV-B) and moves the propagated fraction by an order of magnitude, so that fraction is a property of the graph measured, not a constant of INT8.

Construct. Local mismatch is teacher-forced: it measures an operator’s arithmetic in isolation, and the propagated column is what captures its interaction with upstream error.

Conclusion. The sixteen accuracy comparisons in Tables I and III are not corrected for multiple comparisons; the largest $|\Delta|/SE$ over them is 2.2. Three seeds cover every requantization axis; the rest use one checkpoint per network.

Internal. The external bit-exactness check covers convolution, pooling and fully-connected layers, not depthwise convolution, element-wise addition or softmax.

VI. ARTIFACT AVAILABILITY

Every number is emitted by a script from a committed result file; the scripts, the result files and the training commands are public, and a continuous-integration job regenerates every table here on each commit and fails if a committed one differs. The measurements are `e15_fidelity.py` (Table I, Fig. 1), `e16_ln_emulation.py` (Table III), `e14_rounding_seeds.py` (Fig. 2, Table IV) and `e17_dense_output.py` (Table II); the external check is `e11_tflite_crosscheck.py`. One caveat applies to the seeded measurement: the trainer’s seed controlled data order but not weight initialization when these checkpoints were produced, so the three replicates are independent draws that the same command will not reproduce bit for bit.

VII. CONCLUSION

Simulated quantization predicts INT8 quality well and INT8 tensors badly, on a classifier and on a network whose output is the product, and the two facts are independent. The gap is not carried by any single operator: it is born as ± 1 LSB events in operators that individually disagree on under 1% of their codes and grown by propagation, which is why removing the worst local source recovers only a tenth of it. Teams that need golden vectors should generate them from an integer implementation validated against an external reference, and teams that change a requantization detail should measure it across seeds, because its magnitude — unlike its sign — does not transfer.

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